

Program 6

Docker Image Build, Push to Docker Hub & GitHub Container Registry

1. Create project directory

```
mkdir program-6 && cd program-6
```

Step 1: Create a Dockerfile

```
nano Dockerfile
```

```
FROM alpine:latest
```

```
#a simple command
```

```
CMD ["echo", "Hello Srusti, this is a small image!"]
```

Step 2: Build the Docker Image

```
docker build -t program-6 .
```

Step 3: Tag the Image for Docker Hub

```
docker tag program-6 <dockerhub-username>/program-6:latest
```

Step 4: Push the Image to Docker Hub

Login to Docker Hub using your Docker Hub credentials (Personal Access Token as password).

```
docker login -u <dockerhub-username>
```

Push the image to Docker Hub:

```
docker push <dockerhub-username>/program-6:latest
```

Step 5: Tag the Image for GitHub Container Registry (GHCR)

Tag the same image for GitHub Container Registry.

```
docker tag program-6 ghcr.io/github-username/program-6:latest
```

Step 6: Push the Image to GHCR

Login to GHCR using GitHub Personal Access Token (PAT).

```
echo $GITHUB_PAT | docker login ghcr.io -u <github-username>
```

```
--password-stdin
```

Push the image:

```
docker push ghcr.io/github-username/program-6:latest
```

Step 7: Connect the Local Project to GitHub (Source Code)

```
git init
```

```
git remote add origin https://github.com/<github-username>/program-6.git
```

```
git remote set-url origin
```

```
https://github.com/<github-username>/program-6.git
```

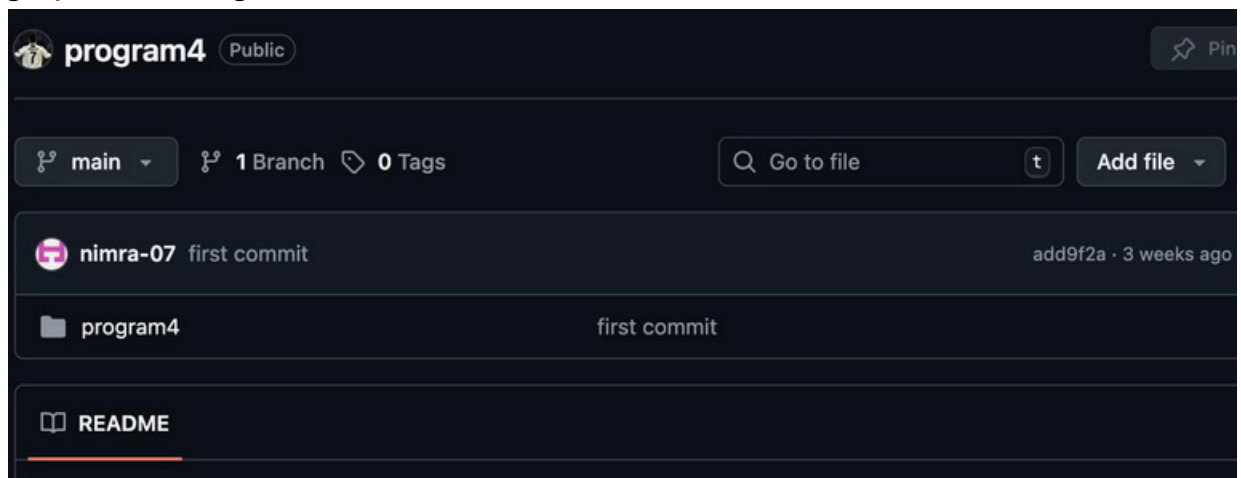
```
git remote -v
```

Step 8: Push the Source Code to GitHub

```
git add .
```

```
git commit -m "Initial commit"
```

```
git push -u origin main
```



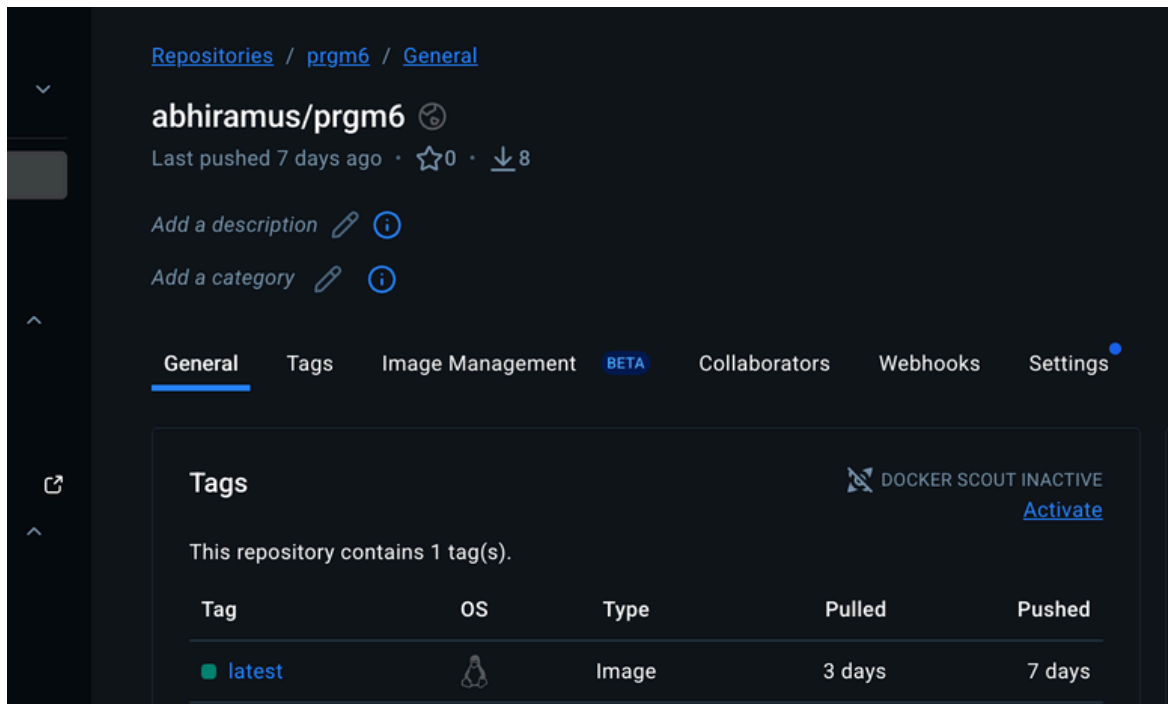
Step 9: Verify Uploads on Both Registries

GitHub

- Ensure the repository program-6 exists on GitHub.
- Navigate to Packages.

Docker Hub

- Login to Docker Hub.
- Go to Repositories.



The screenshot shows the Docker Hub interface for the repository `abhirus/prgm6`. The breadcrumb navigation at the top indicates the path: `Repositories / prgm6 / General`. The repository name `abhirus/prgm6` is displayed with a lock icon, indicating it is private. Below the name, it states "Last pushed 7 days ago" and shows 0 stars and 8 downloads. There are links to "Add a description" and "Add a category", each with an edit icon and an information icon. A horizontal menu at the top of the content area includes "General" (selected), "Tags", "Image Management" (marked BETA), "Collaborators", "Webhooks", and "Settings". The "Tags" tab is active, showing a section titled "Tags" with a "DOCKER SCOUT INACTIVE" warning and an "Activate" link. Below this, it states "This repository contains 1 tag(s)". A table lists the tags with columns for Tag, OS, Type, Pulled, and Pushed.

Tag	OS	Type	Pulled	Pushed
<code>latest</code>	linux	Image	3 days	7 days